BIOL-1 Practice Test 02

Modules 5-6: Membranes and Metabolism

Instructions: This practice test covers material from Modules 5 (Membranes) and 6 (Metabolism). Answer all questions to the best of your ability.

Part A: Multiple Choice

Choose the best answer for each question.

Module 5: Membranes

1. The cell membrane is best described by the:

A) Lock and key model
B) Fluid mosaic model
C) Rigid barrier model
D) Simple lipid model

2. The cell membrane is primarily composed of:

A) Carbohydrates
B) Proteins only
C) Phospholipid bilayer
D) Nucleic acids

3. Phospholipids spontaneously form a bilayer in water because:

A) They are charged molecules
B) They have hydrophilic heads and hydrophobic tails
C) They are completely hydrophobic
D) They are all the same size

4. Cholesterol in the cell membrane functions to:

A) Transport molecules across the membrane
B) Help maintain membrane fluidity
C) Store genetic information
D) Produce ATP

5. Which transport process requires ATP?

A) Diffusion
B) Osmosis
C) Facilitated diffusion
D) Active transport

6. When a cell is placed in a hypotonic solution, water will:

A) Move out of the cell
B) Move into the cell
C) Not move
D) Move in both directions equally

7. A cell in an isotonic solution will experience:

A) Net water movement into the cell
B) Net water movement out of the cell
C) No net water movement
D) Cell lysis

Module 6: Metabolism

8. Enzymes function by:

A) Increasing activation energy
B) Decreasing activation energy
C) Providing energy for reactions
D) Changing the equilibrium of reactions

9. The molecule that stores and transfers energy in all cells is:

A) Glucose
B) ATP
C) DNA
D) NADH

10. The part of an enzyme where the substrate binds is called the:

A) Allosteric site
B) Active site
C) Inhibitor site
D) Product site

11. When an enzyme loses its shape due to heat or pH changes, this is called:

A) Activation
B) Denaturation
C) Inhibition
D) Catalysis

12. ATP releases energy when:

A) A phosphate group is added
B) A phosphate bond is broken (hydrolysis)
C) It binds to DNA
D) It is stored in the cell

13. Which statement about enzymes is TRUE?

A) Enzymes are used up in reactions
B) Enzymes can be reused
C) Enzymes are made of carbohydrates
D) Enzymes increase the activation energy

Part B: Fill in the Blank

Write the correct term in the blank.

14. The phospholipid bilayer is described as a ________ mosaic model.

15. The process by which water moves across a membrane is called ________.

16. Transport that requires ATP is called ________ transport.

17. A solution with a lower solute concentration than the cell is called ________.

18. The specific reactant that an enzyme acts on is called the ________.

19. Energy of motion is referred to as ________ energy.

20. ________ inhibition occurs when a molecule binds to the active site and blocks the substrate.

21. The sum of all chemical reactions in an organism is called ________.

22. Reactions that release energy by breaking down complex molecules are called ________ reactions.

Part C: Short Answer

Answer each question in 2-3 complete sentences.

23. Explain the difference between passive transport and active transport. Give one example of each.

24. Describe what happens to a cell placed in a hypertonic solution. How would this differ for a plant cell versus an animal cell?

25. Explain how enzymes speed up chemical reactions and why they are specific to their substrates.

26. Describe the structure of ATP and explain how it stores and releases energy for the cell.

End of Practice Test 02